Server.py

```
from flask import Flask,request
from tensorflow.keras.models import load_model
import numpy as np

model = load_model('C:\\sharath\projects\\assignments\\week 7\\week7.h5')
model.summary()

label_names = {
0 : 'airplane',
1 : 'automobile',
2 : 'bird',
3 : 'cat',
4 : 'deer',
5 : 'dog',
6 : 'frog',
7 : 'horse',
8 : 'ship',
9 : 'truck',
}

app = Flask(__name__)
@app.route('/')
def welcome():
 return 'hello'
@app.route('/pred',methods = ['POST','GET'])
def serve_model():

  request_data = request.get_json(force=True)
  img = request_data['img']
  img = np.array(img)
  return ('The prediction is {}'.format(label_names[model.predict(img).argmax()]))

if __name__ == '__main__':
 app.run()
```

Running output

```
 * Serving Flask app 'server' (lazy loading)
 * Environment: production
   WARNING: This is a development server. Do not use it in a production deployment.
   Use a production WSGI server instead.
 * Debug mode: off
 * Running on http://127.0.0.1:5000/ (Press CTRL+C to quit)
127.0.0.1 - - [03/Apr/2022 14:12:30] "POST /pred HTTP/1.1" 200 -
```

client.py

```
import numpy as np
import requests
import json
from tensorflow.keras.preprocessing.image import load_img

img = np.array(load_img('C:\\sharath\\projects\\assignments\\week 7\\horse.png'))
img.resize(1,32,32,3)

img = img.tolist()

url = ' http://localhost:5000/pred'
requested_data = json.dumps({'img':img})
headers = {
 'content-type':'application/json'
}

response = requests.post(url,data = requested_data,headers = headers)

print(response.text )
```

Running output

```
python client.py
2022-04-03 14:11:52.275132: W
tensorflow/stream_executor/platform/default/dso_loader.cc:64] Could not load
dynamic library 'cudart64_110.dll'; dlerror: cudart64_110.dll not found
2022-04-03 14:11:52.275328: I tensorflow/stream_executor/cuda/cudart_stub.cc:29]
Ignore above cudart dlerror if you do not have
a GPU set up on your machine.
The prediction is cat
```